

Chapter 1

Snow products

1.1 IT-Snow

IT-SNOW is a high resolution, multi-year snow reanalysis dataset covering the whole Italian territory. It provides daily maps of snow water equivalent (SWE), snow depth, bulk snow density and liquid water content of the snowpack for a period spanning from September 1st, 2010 to August 31st, 2025. The spatial resolution is approximately 500 meters.

1.1.1 S3M

S3M, also known as *Snow Multidata Mapping and Modelling*, is an hydrology-oriented cryospheric model that simulates seasonal snow evolution through time. S3M is a raster-based model treating pixels as independent units, as it neglects mass and energy transport processes such as wind redistribution. The same set of ordinary differential equations (ODEs) is solved for each grid point by means of the Euler method: the hourly timestep can be adjusted by the user to meet specific application needs.

